Conjecture 00000005045: local Hanf data do not give global isomorphism

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23 October 2026

1 The necessary transfer fails at degree two

The source explicitly defines the local-to-global conversion as conversion from Hanf local isomorphism to global isomorphism, and asserts a uniform radius-two conversion constant table governed by degree thresholds. Under that stated definition, the necessary transfer already fails for finite graphs of degree two. For each integer $t \geq 0$, put

$$G_t = (t+1)C_{12}, \qquad H_t = 2(t+1)C_6,$$

where multiplication denotes disjoint copies of the actual cycle graph. Both graphs have $12(t+1)$ vertices, and every vertex has degree 2. Every rooted induced closed radius-two ball in either graph is the same five-vertex path, rooted at its middle vertex. Consequently the graphs have a bijection of vertices preserving rooted radius-two neighborhood types. In fact, the number of roots of every neighborhood type is exactly equal, so every finite Hanf count threshold agrees. Yet G_t and H_t are not isomorphic. Taking t arbitrarily large excludes any size cutoff as well.

We do not assign numbers to the source's unspecified constant table. Rather, any uniform radius-two threshold/size-cutoff rule that accomplishes the stated conversion must work on this family, and cannot do so. A table for a different, unstated operation is outside this conclusion. No independent claim about numerical entries 3 or 5, monotonicity, or a size-doubling jump is made.

2 Actual balls, isomorphisms and Hanf counts

Lean uses Mathlib's actual `SimpleGraph.cycleGraph`. Disjoint copies have vertices (i, v), with adjacency precisely equality of copy indices and actual cycle adjacency. The closed radius-two ball is defined by

$$B_2(v) = \{w : w = v \ \vee \ v \sim w \ \vee \ (\exists z, \ v \sim z \ \wedge \ z \sim w)\}.$$

The proof formally identifies this with existence of an actual graph walk of length at most 2. It does not use the default value of a distance between disconnected vertices. A rooted ball isomorphism is a genuine graph isomorphism of the induced subgraphs, carrying the actual root to the other root.

At root 0, the vertices of these balls are respectively $[0, 1, 2, 10, 11]$ in C_{12} and $[0, 1, 2, 4, 5]$ in C_6. The proof checks their actual adjacency relations, constructs the induced graph isomorphism, and then uses genuine cycle rotations and copy projections for arbitrary roots. Ordinary kernel-checked finite decisions are used for the small base graphs.

For any finite ambient graph A and any rooted target graph (K, z), the number of roots whose induced ball is isomorphic to $B_2^K(z)$ is the cardinality of that actual subtype of vertices. A Hanf vertex bijection gives an explicit equivalence between these subtypes in G_t and H_t. Thus their finite cardinalities are equal for every target, not merely for an assigned list of counts. In particular, for every threshold q, the counts of each finite rooted type are equal or both exceed q, with equality always supplying the first alternative. `Nat.card` is used only on these finite ambient vertex subtypes; no infinite-cardinality default is invoked.

3 Global nonisomorphism and arbitrary size

Every actual walk in a disjoint-copy graph preserves the copy index. The actual twelve-cycle is connected. If a global graph isomorphism existed, the images of all twelve vertices of one C_{12} component would therefore lie in one C_6 component. Injectivity of the graph isomorphism would induce an injection $\mathrm{Fin}\,12 \hookrightarrow \mathrm{Fin}\,6$, contradicting $12 > 6$. This proves nonisomorphism for every t.

Lean proves the actual orders and all actual vertex degrees. Its final theorem states that for every size cutoff N and every count threshold q, some t gives equal-order graphs of order at least N, degree exactly 2, genuine Hanf matching and threshold agreement, but no global isomorphism. The formal negation of a uniform cutoff/threshold conversion follows.

4 Scope, references and verification

Hanf locality in its usual form yields bounded-quantifier-rank first-order equivalence, not global graph isomorphism. This submission does not refute that theorem. The distinction is explicit in Anuj Dawar's primary Simons Institute lectures: *Finite and Algorithmic Model Theory II: Automata-Based Methods*, 30 August 2016, PDF page 4, gives the neighborhood bijection definition; lecture III, *Parameterized Satisfaction*, 31 August 2016, PDF pages 9–10, gives induced neighborhoods, threshold counts and the first-order conclusion. The directly read sources are lecture II and lecture III.

The package pins Lean 4.33.0 and Mathlib commit

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